

Solution Brief

Structured Deadline Extraction and Calendar Integration for Moodle

Executive Summary

University students must manage deadlines across multiple communication channels. While some appear in the Moodle calendar, many more are only mentioned in PDFs, such as syllabuses and assignment descriptions, or in SharePoint document and HTML-based room and exam systems. Consequently, important academic deadlines remain fragmented and partially invisible. Based on observations of courses, it is estimated that around 50% of relevant deadlines are not fully represented in the Moodle calendar.

Syllabus-Sync addresses this issue by converting unstructured course information into structured calendar events. Students upload course documents, relevant deadlines are identified and normalised, and structured events can be exported into personal calendars.

This solution operates in a privacy-conscious and user-controlled manner and does not require system-level access to Moodle or email accounts.

Business Value

- **Improved transparency:** Deadlines embedded in documents are made visible and easier to manage.
- **Reduced administrative friction:** Students no longer need to manually monitor multiple systems.
- **Lower error rate:** Structured calendar entries reduce the number of missed or late submissions.
- **Process standardisation:** Different communication channels are consolidated into one consistent output format.
- **Quantifiable impact:** Missed deadlines caused by incomplete calendar visibility are reduced.
- **Operational benefit for lecturers:** With consent, structured deadline data can be forwarded to secretariats, reducing the need for repetitive clarification requests and administrative follow-up.

- **Return on investment:** Even avoiding a single missed submission per semester would be valuable for both students and the academic administration.

Use Case Description

Users:

- **Primary:** University students
- **Secondary:** Academic staff (indirect)

Beneficiaries:

- Students (planning reliability)
- Academic programmes (reduced late submissions)
- Administrative units (fewer deadline-related queries)

As-is-process:

Deadlines are distributed across:

- Moodle assignments
- PDF syllabuses
- HTML pages
- SharePoint folders (e.g. exam schedules)
- Room reservation tools

Students must consolidate this information manually and create their own calendar entries. Deadlines that are mentioned only in documents often do not appear in the Moodle calendar.

To-Be Process:

1. The user uploads the relevant course documents
2. Deadlines and task types are then identified and standardised
3. The user reviews the extracted events
4. Structured calendar entries are then exported in .ics format
5. The user then imports these entries into a personal calendar

Functional scope

Must-have:

- Upload course documents (PDF, Word, HTML, Excel)
- Identify deadlines (date, time, task, description)
- Multi-language support (German, English, French, Italian and Spanish)
- Date format normalisation (e.g. “12th January 2026”, “12.01.2026”, January 12, 2026)
- Categorise task type (exam, assignment, presentation)
- iCal (.ics) export and import
- Follow-up reminders
- Time zone handling
- User confirmation before export

Could-Have:

- Extraction of course sections, rooms and lecture times
- Chat between students, lecturers and administration
- Automated synchronisation
- Data selection and filtering options
- Room booking verification
- Holiday and block week detection
- Extended “AI tutor” functionality

Not-Have:

- Direct login to Moodle or email
- Persistent data storage (no database)
- Unlimited historical data retention
- Automatic background data access
- Full write access to external calendar systems

Design Principle: Minimal system integration, maximum user control

Feasibility

- **Technical:** It is feasible to use established text-processing methods for document parsing and structured event generation.
- **Operational:** A user review step ensures quality control.
- **Organisational:** The solution does not interfere with existing Moodle or SharePoint systems.
- **Alternatives:** Relying solely on the Moodle calendar is insufficient, given that a significant proportion of deadlines are communicated exclusively through documents or external systems, such as SharePoint and HTML-based room reservation tools.

A purely rule- or regex-based extraction approach was also considered. However, due to the variety of formatting, multilingual content and implicit deadline descriptions, a flexible text interpretation approach is more reliable.

Although manual consolidation is possible, it is inefficient, time-consuming and error-prone.

Data requirements

- **Type:** Unstructured academic documents (PDF, HTML, Word and Excel).
- **Source:** User-uploaded course materials from Moodle, SharePoint or similar systems.
- **Quality:** Inconsistent formatting and multilingual content.
- **Ownership:** Documents remain user-owned and are only processed temporarily.

Architecture Proposal

High-Level Architecture:

- File upload
- Text extraction
- Deadline Identification
- Event Structuring
- User Review
- Calendar export (iCal)

Design Characteristics:

- Stateless processing
- No database
- Human in the loop validation
- Output standardisation via iCal format

Technologies:

- Document parsing and text extraction tools for PDF, Word, HTML, and Excel files
- Generative AI model for deadline identification and text interpretation
- JSON schema for structured event representation
- iCalendar (.ics) standard for cross-platform calendar export

Implementation Roadmap

- **Initiation:** Define the scope, supported formats and event structure.
- **Development:** Implement the document upload, extraction and validation interface, and export.
- **Deployment:** Test the prototype using real Moodle documents.
- **Optimisation:** Improve accuracy and expand format compatibility.

Success Criteria

- ≥85% correct deadline identification in test cases
- Less than 2 minutes required to import deadlines per course
- Zero automatic calendar modifications without user approval

Risks & Concerns

Risk	Metric	Countermeasure
Incorrect deadline identification	Error rate after user review	Mandatory user confirmation
Ambiguous date formats	Number of flagged cases	Standardisation and user review
Data privacy concerns	Data retention duration	Stateless processing, no storage

Open Discussion Points

- Should reminder timing be configurable?
- Should structure data optionally be shared with administrative units?
- How should conflicts between multiple document sources be resolved?
- Is the .ics format sufficient, or is direct calendar API integration required?
- Should exam schedules be handled differently from assignments?
- Should lecturers be able to approve the forwarding of extracted deadlines to departmental secretariats?
- How should conflicts between multiple data sources (Moodle, SharePoint and room systems) be prioritised?

References

- Google Skills - Generative AI Learning Path
- iCalendar Specification
- Google Antigravity Documentation